EM exercise 01.m | EM exercise 02.m

```
# IT26102084 - Kuruvitage.S.P.M
printf("IE1014 : Engineering Mathematics - I\n\n")
printf("*************   Lab Sheet - 1.1   *************\n\n")

2^10            # a
sqrt(10)
exp(10)
log(10)

ceil(2.25)      # b
fix(2.25)
round(2.25)
floor(2.27)

sin(1)          # c
cosd(90)
asin(0.5)
atan(-1)

quotient = idivide(int32(9), int32(4));   # d
remainder = mod(9, 4);
printf("\nQuotient of 9 devided by 4 = %d\n", quotient)
printf("\nRemainder of 9 devided by 4 = %d\n\n", remainder)

M = [9.2, 40 ; 5^2, 3];         # e
printf("Determinant:\n")
disp(det(M))
printf("\nInverse:\n")
disp(inv(M))
printf("\nEigen:\n")
disp(eig(M))
```

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octave:14> IE1014 : Engineering Mathematics - I

*************   Lab Sheet - 1.1   *************

ans = 1024
ans = 3.1623
ans = 2.2026e+04
ans = 2.3026
ans = 3
ans = 2
ans = 2
ans = 2
ans = 0.8415
ans = 0
ans = 0.5236
ans = -0.7854

Quotient of 9 devided by 4 = 2

Remainder of 9 devided by 4 = 1

Determinant:
-972.40

Inverse:
  -3.0852e-03   4.1135e-02
   2.5710e-02  -9.4611e-03

Eigen:
   37.874
  -25.674
```

EM exercise 01.m   EM exercise 02.m

```
# IT26102084 - Kuruvitage.S.P.M
printf("IE1014 : Engineering Mathematics - I\n\n")
printf("*************    Lab Sheet - 1.2    *************\n\n")

a = 10;
b = 5;

c1 = a + b ;
c2 = a - b ;
c3 = 10*a + 5*b ;
c4 = a / b ;
c5 = a ^ b ;
printf('c1=%.2f\n c2=%.2f\n c3=%.2f\n c4=%.2f\n c5=%2f\n', c1, c2, c3, c4, c5)

# x^5 + 3x^4 + 2x^3 + 9x^2 + 4x = 0

poly = [1 ,3 ,2 ,9 ,4 ,0 ];
r = roots(poly);
printf("\n Roots of the polynomial are :\n\n");
disp(r)

```

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*************    Lab Sheet - 1.2    *************

c1=15.00
 c2=5.00
 c3=125.00
 c4=2.00
 c5=100000.000000

 Roots of the polynomial are :

  -3.1453 +      0i
   0.3047 + 1.6270i
   0.3047 - 1.6270i
  -0.4641 +      0i
        0 +      0i
```